Supplementary Materials

AgenticSum: An Agentic Framework for
Faithful Clinical Summarization

Note: Cross-references of the form "Table X" or "Figure X" without an explicit "(main paper)" qualifier refer to the main submission. Appendix sections are self-contained and reproducible without the main paper.

A Algorithm: AgenticSum

This appendix provides a formal algorithmic specification of `AgenticSum`, detailing the supervised agentic inference procedure used for clinical summarization. The algorithm explicitly delineates the roles of each agent and the control flow governing iterative refinement, complementing the high-level framework description in Section III.

Algorithm 1 summarizes the end-to-end inference process. Given a clinical document D, the *FocusAgent* first performs deterministic sentence-level input compression using attention-based salience estimates, producing a reduced context D_{reduced}. The *DraftAgent* then generates an initial summary $S^{(0)}$ conditioned solely on the compressed input.

Subsequent refinement proceeds through a supervised loop. At each iteration, the *HallucinationDetectorAgent* evaluates the current summary by combining attention-to-underlying-reference alignment (AURA) scores with entailment-based verification to identify unsupported spans. The *FixAgent* selectively revises only the spans flagged as hallucinated, preserving supported content and avoiding full regeneration. Refinement is governed by the *ClinicalSupervisorAgent*, which monitors convergence via changes in aggregate grounding statistics and enforces a bounded refinement budget.

B Sensitivity of AURA to the Grounding Threshold

We examine the sensitivity of AURA to the grounding threshold τ, which determines whether a generated span is considered sufficiently supported by the source document. This parameter governs the balance between accepting weakly supported content and triggering corrective refinement. Rather than emphasizing absolute performance values, this analysis focuses on the stability of the system's behavior across a range of thresholds.

The threshold $\tau = 0.42$ is selected to occupy a principled position between two failure modes: values that are too permissive allow weakly supported spans to pass uncorrected, reducing the grounding signal's discriminative utility, while values that are too strict trigger excessive intervention on spans that are sufficiently grounded, introducing over-correction artifacts. At $\tau = 0.42$, the grounding criterion is calibrated to the empirical distribution of AURA scores observed on development data, where the majority of well-supported spans cluster above this boundary and hallucinated spans fall below it. This selection reflects the operational semantics of the threshold as a decision boundary rather than a tuning parameter, and is consistent with the broader observation in attention-based grounding literature that moderate thresholds over normalized scores yield stable separation without sensitivity to exact numerical choice [1].

C Sensitivity of AURA to Attention Layer and Head Aggregation

We further analyze the sensitivity of AURA to the choice of attention layer and head aggregation strategy used to derive grounding signals. Since transformer representations vary across depth, different layers may

Algorithm 1: AgenticSum: Supervised Agentic Inference for Faithful Clinical Summarization

Input: Clinical document D, model parameters θ, retention ratio r, grounding threshold τ, tolerance ϵ, max steps $T_{\max}$

Output: Final grounded summary S^*

`// FocusAgent: Input Compression`

Partition D into sentences $\{u_1, \dots, u_m\}$

Run single forward pass over D; extract last-layer attention $A^{(L)} \in \mathbb{R}^{H \times T \times T}$

foreach *head h, token k* **do**

  $C_{h,k} \leftarrow \sum_{i \geq k} A^{(L)}_{h,i,k}$ `// column sum from subsequent positions`

foreach *sentence u_j with token indices T_j* **do**

  $\beta_j \leftarrow \dfrac{1}{H\,|T_j|} \sum_{h=1}^{H} \sum_{k \in T_j} C_{h,k}$

Select top-$k = \lfloor r \cdot m \rfloor$ sentences ranked by β_j

$D_{\text{reduced}} \leftarrow$ selected sentences

`// DraftAgent: Initial Generation`

$S^{(0)} \leftarrow \text{Generate}(D_{\text{reduced}}; \theta)$

Compute span set $\{z_1, \dots, z_M\}$ from $S^{(0)}$

foreach *span z_j* **do**

  Compute grounding score a_j

  Compute hallucination label h_j

$\bar{A}^{(0)} \leftarrow \frac{1}{M} \sum_{j=1}^{M} a_j$

$\mathcal{H}^{(0)} \leftarrow \{z_j \mid h_j = 1 \text{ or } a_j < \tau\}$

for $t \leftarrow 0$ **to** $T_{\max} - 1$ **do**

  if $\mathcal{H}^{(t)} = \emptyset$ **then**

    return $S^{(t)}$

  `// FixAgent: Targeted Revision`

  $S^{(t+1)} \leftarrow \text{Fix}(D_{\text{reduced}}, S^{(t)}, \mathcal{H}^{(t)}; \theta)$

  Segment $S^{(t+1)}$ into spans $\{z_1, \dots, z_M\}$

  foreach *span z_j* **do**

    Compute a_j, h_j

  $\bar{A}^{(t+1)} \leftarrow \frac{1}{M} \sum_{j=1}^{M} a_j$

  $\mathcal{H}^{(t+1)} \leftarrow \{z_j \mid h_j = 1 \text{ or } a_j < \tau\}$

  if $|\bar{A}^{(t+1)} - \bar{A}^{(t)}| < \epsilon$ **then**

    return $S^{(t+1)}$

  if $\mathcal{H}^{(t+1)} \setminus \mathcal{H}^{(t)} = \emptyset$ **then**

    return $S^{(t+1)}$

return $S^{(T_{\max})}$

encode distinct aspects of the relationship between generated and source text. Similarly, aggregation across attention heads may influence how grounding signals are combined.

The choice of final-layer attention with mean aggregation across heads is motivated by two complementary considerations. First, final-layer representations in autoregressive decoders encode the most contextualized token-level information, reflecting the model's cumulative integration of source content at the point of generation [1]. This makes final-layer attention a principled signal for assessing whether a generated token is grounded in the source rather than produced from parametric knowledge. Second, mean pooling across heads provides a statistically stable aggregation that avoids the sparsity artifacts introduced by max pooling, where a single outlier head can dominate the grounding estimate. Together, this configuration minimizes sensitivity to layer-specific representational idiosyncrasies and head-level variance, yielding a grounding signal that reflects distributed model behavior rather than localized activation patterns.

D Human Evaluation and Statistical Analysis

This appendix provides additional details regarding the human evaluation protocol and statistical analyses summarized in Table 3. Each participant evaluated three clinical note scenarios spanning psychiatric, cardiopulmonary, and oncology domains. For each scenario, participants were presented with two summaries: one generated by a vanilla large language model (LLM) and one produced by AgenticSum. Presentation order was randomized to mitigate ordering effects. Participants were asked to (i) identify which summary contained more hallucinated content (forced-choice preference task), and (ii) independently rate the hallucination severity of each summary on a five-point ordinal scale, where lower values indicate fewer hallucinations.

Hallucination Severity and Paired Testing. Hallucination severity was measured on an ordinal scale ranging from 1 (no hallucinations; all content supported by the original note) to 5 (severe hallucinations; extensive fabricated or contradictory information). Let $H_i^{(V)}$ and $H_i^{(A)}$ denote the severity ratings assigned by participant i to the vanilla LLM summary and the AgenticSum summary, respectively. Because ratings are ordinal and paired at the participant level, differences in hallucination severity were assessed using the paired Wilcoxon signed-rank test [2]. Specifically, we evaluated whether the median of the paired differences

$$\Delta_i = H_i^{(V)} - H_i^{(A)}$$

was significantly greater than zero, indicating reduced hallucination severity for AgenticSum. Statistical significance was evaluated at $\alpha = 0.05$.

Effect Size and Preference Dominance Testing. In addition to significance testing, we report rank-biserial correlation (r_{rb}) as an effect size measure appropriate for paired ordinal data [3], defined as

$$r_{\mathrm{rb}} = \frac{n_+ - n_-}{n_+ + n_-},$$

where n_+ and n_- denote the number of positive and negative paired differences, respectively. Observed effect sizes across all clinical domains were large ($r_{\mathrm{rb}} \in [0.80, 0.91]$).

To assess annotator preference consistency, we conducted preference dominance tests using a binomial test. Correct-guess accuracy reflects the proportion of participants who identified the same summary as more hallucinated within a given clinical domain. Dominance tests evaluated whether the observed preference rate for AgenticSum exceeded chance under a null hypothesis of random selection (probability $p = 0.5$). Dominance p-values reported in Table 3 indicate that annotator preferences for AgenticSum were significantly above chance across all domains.

E Hallucination Annotation Prompt

To identify hallucinated spans in model-generated clinical summaries, we employ a structured evaluation prompt grounded in strict textual entailment. The task requires a language model to verify whether each statement in a generated summary is **verifiably supported** by the source document, meaning the information must be *explicitly stated and unambiguous* in the document text. This prompt is adapted from the Lookback Lens [1] framework, but incorporates additional guardrails specific to clinical applications. Crucially, it includes explicit instructions to prevent confirmation bias [4], discourage plausible sounding inferences, and eliminate reliance on domain knowledge not grounded in the text.

> You will be provided with:
>
> - A clinical source document
> - A machine-generated summary
>
> Your task is to assess whether the summary is **strictly entailed** by the source document.
> **Definition of Entailment:** A statement is entailed only if it is explicitly and unambiguously stated in the document. Any statement that requires inference, clinical reasoning, or prior medical knowledge should be treated as **Not Entailed**.
> **Key Instructions:**
>
> - DO NOT infer or assume missing information.
> - DO NOT use clinical knowledge, reasoning, or common sense.
> - DO NOT validate based on plausibility, typicality, or tone.
> - DO NOT guess what the author likely meant.
>
> You must treat the document as the **only source of ground truth**.
> **Document:** {document}
> **Proposed Summary:** {summary}
> **Output Format:**
>
> - **Entailment Label:** `Entailed` or `Not Entailed`
> - **Explanation:** Justify your label with direct references to the document.
> - **Problematic Spans (if any):** List the summary phrases not directly supported by the source.

Example Output:

> **Entailment Label:** `Not Entailed`
> **Explanation:** The summary states "Patient has a history of diabetes," but the document does not mention diabetes explicitly. No evidence is found.
> **Problematic Spans:** ["history of diabetes"]

F LLM-as-a-Judge Evaluation Prompt

The following instruction prompt is used to evaluate generated clinical summaries against their corresponding source documents using an instruction-tuned large language model acting in an evaluative role. The model is instructed to assess summaries strictly with respect to the provided source document, without relying on external medical knowledge or plausible inference.

Task: Evaluate the generated medical summary against the source clinical document. Assign an integer score from 1 to 5 for each criterion defined below.

Source Document:

`{SOURCE_DOCUMENT}`

Generated Summary:

`{GENERATED_SUMMARY}`

Evaluation Criteria (1–5 scale):

- **Hallucination:** Degree of unsupported or fabricated content (1 = no hallucination; 5 = major fabrications)
- **Factual Consistency:** Faithfulness of statements to the source document (1 = highly inaccurate; 5 = fully accurate)
- **Completeness:** Coverage of core clinical information (1 = key information missing; 5 = fully comprehensive)
- **Coherence:** Fluency and logical organization of the summary (1 = poorly written; 5 = highly coherent)

Output Format: Return the scores using the following strict key–value format, with no additional text:

```
Hallucination: X
Factual: X
Complete: X
Coherent: X
```

Evaluation uses a low decoding temperature (0.3) and a maximum output length of 150 tokens; all scores are validated to lie within the 1–5 range, yielding a reproducible and fully automated large-scale assessment protocol.

G Hallucination Detection Example

We present a representative hallucination annotation example generated by our Hallucination Detector agent module. The summary is evaluated using our factuality prompt protocol (see Appendix E), which flags spans not explicitly supported by the source document.

H FocusAgent: Sentence-Level Filtering Example

We present a representative example demonstrating how FocusAgent performs sentence-level filtering to produce a concise and clinically coherent subset of the input document. This example is drawn from the MIMIC-IV dataset and illustrates the type of input compression used to guide downstream summarization. Table 2 shows the retained and discarded sentences alongside their computed salience scores.

I Full Pipeline Example

We demonstrate the full AgenticSum pipeline using a real-world clinical note. This example illustrates how hallucinations introduced during draft generation by DraftAgent are detected and corrected by FixAgent. Table 3 presents the source document, initial draft, detected spans, and final corrected output.

J Qualitative Error Analysis

To better understand the limitations of `AgenticSum`, we analyze a representative failure case from held-out samples. Note 16002318-DS-17 from the MIMIC-IV dataset is a post-operative discharge summary for a

Source Document (Excerpt)
Patient was in good health until 10 days ago, when he began experiencing gradually worsening headaches. Head CT revealed an intracranial mass lesion. An enhancing focus in the left parapharyngeal region likely represents an abnormally enlarged lymph node, though it is not fully visualized. The imaging raises the possibility of a benign neoplasm (question neurogenic tumor), and pathology is pending. Discharge medications include atenolol, phenytoin, docusate sodium, and dexamethasone.
Generated Summary
The patient is a *10-year-old male* who was experiencing worsening headaches. A CT scan revealed an intracranial mass lesion, which was found to be *an abnormally enlarged lymph node representing metastatic involvement*. The *pathology pending at the time of discharge is a benign neurogenic tumor*. The patient was prescribed atenolol, phenytoin, docusate sodium, and dexamethasone.
Hallucination Detector agent Analysis
Factuality Verdict: UNSUPPORTED **Explanation:** The patient's age and gender are unspecified. The lymph node is described as potentially benign, not definitively metastatic. Pathology is still pending at discharge. The hallucinated spans introduce unwarranted specificity and certainty. **Detected Hallucinations:** - `"10-year-old male"` — Demographics not provided - `"representing metastatic involvement"` — No mention of metastasis - `"pathology...benign neurogenic tumor"` — Pathology result is pending **Error Types:** Demographic fabrication, diagnostic certainty inflation, temporal misalignment

Table 1: Example of hallucination detection using Hallucination Detector agent with span-level grounding analysis.

female patient with morbid obesity (BMI of 51) admitted for laparoscopic gastric bypass surgery. The clinical course was uneventful: the patient remained afebrile with stable vital signs, tolerated advancement to a bariatric diet, and was discharged home in good condition. The reference summary reflects this trajectory concisely, focusing on the procedure, absence of complications, and discharge status.

`AgenticSum` correctly captures the core clinical course, including the patient's stable condition, pain management, and discharge outcome, but additionally incorporates content from discharge instructions. The generated output includes: *"the patient was advised to avoid heavy lifting and abdominal exercises for 6 weeks,"* along with references to wound care, dietary regimen, and follow-up guidance. A secondary imprecision is observed in the procedure description, which is generalized as "lower abdominal surgery" rather than laparoscopic gastric bypass. These spans are not flagged by AURA and persist through FixAgent and the ClinicalSupervisorAgent.

Analysis of the source document shows that semantically similar instructions are present, with similarity scores exceeding the grounding threshold τ, leading AURA to classify these spans as grounded. However, two limitations arise. First, the source instruction contains a redacted weight threshold ("No heavy lifting of items [REDACTED] pounds for 6 weeks"), resulting in reduced specificity in the generated summary. Second, the included content originates from discharge instructions rather than the clinical course, whereas clinical summaries typically prioritize medically salient events. This example highlights a limitation that is not attributable to factual hallucination. The generated content is supported by the source document but is not aligned with the intended scope of the summary. This reflects a distinction between grounding fidelity and summarization relevance. While the current framework emphasizes faithful attribution, it does not explicitly model section-level importance or clinical prioritization. Addressing this limitation would require incorporating document structure awareness or task-specific summarization constraints alongside grounding,

Source Document (Clinical Note)
<SEX> F
<SERVICE> SURGERY
<ALLERGIES> Iodine / contrast dye allergy.
The patient reports persistent abdominal pain that has worsened over the past week.
She states that despite taking her medications, the pain has not resolved.
She denies fever, chills, cough, or shortness of breath.
She has experienced episodes of vomiting but denies diarrhea.
She has a history of GERD and has been taking omeprazole regularly.
Her labs were reviewed and remain stable.
She is alert, oriented, and ambulatory without assistance.
FocusAgent Output (Compressed Input)
[...]
The patient reports persistent abdominal pain that has worsened over the past week.
She states that despite taking her medications, the pain has not resolved.
She denies fever, chills, cough, or shortness of breath.
She has experienced episodes of vomiting but denies diarrhea.
She has a history of GERD and has been taking omeprazole regularly.
Her labs were reviewed and remain stable.
DraftAgent Summary (Generated from Compressed Input)
The patient reports worsening abdominal pain despite medication use. She denies fever, chills, cough, or shortness of breath. She has experienced episodes of vomiting but denies diarrhea. Her history is notable for GERD, and she continues to take omeprazole. Laboratory results were reviewed and found to be stable. She remains alert, oriented, and ambulatory.

Table 2: Example of sentence-level input filtering performed by FocusAgent, illustrating how clinically informative sentences are retained while low-information content is omitted.

which we consider an important direction for future work.

K Clinical Context

K.1 Joint Commission Standard IM.6.10 (EP 7)

In this work, IM.6.10 (EP 7) is cited as a motivating example of supervisory authorization in discharge documentation workflows. Our method adopts the workflow analogue through an explicit supervisory release gate that determines when iterative refinement terminates, without treating the standard as an algorithmic specification or as a guarantee of content correctness. The ClinicalSupervisorAgent therefore governs termination and release of the automated draft, rather than regulatory compliance or clinical validation. Joint Commission Standard IM.6.10 (Element of Performance 7) concerns the completion and authentication of discharge summaries in accredited hospital settings. It requires that discharge summaries be completed and authenticated by the responsible practitioner within a time frame defined by institutional policy, supporting accountability, continuity of care, and patient safety during transitions of care. The standard is primarily applicable to inpatient discharge documentation and does not prescribe uniform requirements for other note types, which are typically governed by local institutional and payer policies. IM.6.10 (EP 7) specifies expectations for documentation workflows rather than prescribing how documentation content is produced. In practice, discharge summaries may be drafted by trainees or non-physician practitioners but are reviewed and authenticated by the responsible clinician prior to finalization. The standard therefore emphasizes supervisory sign-off and responsibility rather than automated verification or correctness guarantees.

K.2 Use Case: Clinical Handoff Documentation

Clinical handoffs (handover) are structured transitions in which patient information and clinical responsibility are transferred between providers during shift changes, unit transfers, or discharge. These transitions often require rapid synthesis of complex EHR data under time pressure, which motivates summarization systems that prioritize factual consistency and traceability to the source record.

We include clinical handoff documentation as a motivating use case for AgenticSum because it emphasizes two requirements that are central to our framework. First, the summary must remain faithful to documented evidence, as unsupported statements can plausibly mislead downstream decision-making during care transitions. Second, the summarization process benefits from explicit control signals that indicate when content is weakly supported and when refinement should stop. AgenticSum addresses these requirements through a staged inference-time procedure that separates drafting from verification and targeted repair. Given an input document D, the system first performs input compression to retain handoff-relevant context, generates an initial draft summary, evaluates the draft using the grounding signal defined in the main method (AURA) to identify weakly supported content, and then applies targeted revision restricted to the flagged spans. A supervisory release gate terminates refinement when stabilization criteria indicate diminishing returns or when a fixed iteration budget is reached, consistent with the termination rule described in the Methodology. In this setting, the output of the framework can be interpreted not only as a finalized handoff summary but also as an auditable refinement trace consisting of flagged spans and their corresponding revisions.

References

[1] Y.-S. Chuang, L. Qiu, C.-Y. Hsieh, R. Krishna, Y. Kim, and J. Glass, "Lookback lens: Detecting and mitigating contextual hallucinations in large language models using only attention maps," *arXiv preprint arXiv:2407.07071*, 2024.

[2] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[3] N. Cliff, "Dominance statistics: Ordinal analyses to answer ordinal questions," *Psychological Bulletin*, vol. 114, no. 3, pp. 494–509, 1993.

[4] F. I. Adiba, Y. Zhang, and R. Beheshti, "Bias and Fairness in Medical LLMs: An Extensive Scoping Review," Jun. 2025. [Online]. Available: https://osf.io/fqejh_v1/

Source Document (Clinical Note)
Male patient with major depressive disorder presents with chronic, intractable nausea that began in childhood and is associated with his depression. Last night, he experienced the worst nausea of his life, leading to suicidal ideation with a plan for carbon monoxide poisoning. He is currently undergoing ECT treatment (2 sessions completed). The patient reports anxiety exacerbates his nausea, and he describes himself as a "worrier." He no longer has active suicidal thoughts but worries about future episodes. Physical exam shows psychomotor retardation, depressed affect, slow speech with pauses, and fine tremor. Lab results are within normal limits. He is alert, oriented, and ambulatory. Current medications include: omeprazole, ranitidine, phenelzine, clonazepam, tamsulosin, and aripiprazole.
DraftAgent Summary (Initial)
The patient is a male who has been experiencing intractable nausea associated with depressive symptoms. He has *a history of car accidents* and has been prescribed ECT treatments. The patient reports that he is *upset about his inability to throw up* and is experiencing anxiety. He has a history of depression and anxiety, and has been taking medication for both conditions. The patient's plan is to try the first round of ECT, and he is *being treated by Dr. Stern*. The patient's *physical examination reveals normal findings*. The patient's speech is slow and with many pauses, and his behavior is cooperative with psychomotor retardation. The patient's thought process is linear and goal-directed. The patient is currently taking several medications, including omeprazole, ranitidine, phenelzine, clonazepam, tamsulosin, and aripiprazole. The patient is alert and interactive, and is ambulatory and independent.
FixAgent Summary (Corrected)
The patient is a male with major depressive disorder and persistent nausea. He has undergone two ECT sessions. The patient reports distress due to the intractable nausea, which is worsened by anxiety. He is undergoing ECT under medical supervision. The physical exam shows psychomotor retardation, slow speech, and cooperative behavior. The patient's thought process is linear and goal-directed. He is currently taking omeprazole, ranitidine, phenelzine, clonazepam, tamsulosin, and aripiprazole. He is alert, interactive, and ambulatory.
FixAgent Analysis
Detected Hallucinations: 4 total **Corrections:** - `"history of car accidents"` — **REMOVED**: Not supported by source - `"inability to throw up"` — **REFINED**: Rephrased as "inability to alleviate nausea" - `"being treated by Dr. Stern"` — **REMOVED**: No provider name in source - `"normal findings"` — **CORRECTED**: Replaced with actual clinical observations **Improvement Metrics:** Hallucination score: 4.2 → 1.8; Factual consistency: 2.1 → 4.3

Table 3: End-to-end example of hallucination detection and correction using the AgenticSum pipeline.